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“Utah Intracortical Electrode Array: Bridging Neural Intent and Machine Response”

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Abstract

The Utah Electrode Array (UEA) is an invasive intracortical microelectrode system designed for high-resolution recording and, in experimental settings, electrical microstimulation of cortical neurons (Maynard et al., 1997; Rousche & Normann, 1999). Composed of a 10×10 grid of penetrating electrodes, the UEA enables single-unit and multi-unit neural activity measurements, providing precise access to neuronal firing patterns that are critical for experimental neuroscience and brain-computer interface (BCI) applications (Rousche & Normann, 1998; Kim et al., 2009).

This paper reviews the UEA's design, functional principles, chronic recording performance, and material- and tissue-related limitations. Evidence is drawn from animal implantation studies, human cortical implants, and explant analyses, highlighting both the array's capabilities and the challenges associated with long-term use (Rousche & Normann, 1998; Woeppel et al., 2021). Despite gradual declines in signal quality due to biological responses such as glial scarring and fibrotic encapsulation, and material degradation including coating delamination and insulation failure, the UEA remains a foundational tool for neural interfacing and serves as a benchmark for the development of next-generation intracortical arrays (Kim et al., 2009; Woeppel et al., 2021).

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I. Introduction

Neural activity is the fundamental language through which the brain perceives, decides, and acts. When this system is disrupted by traumatic injury, neurodegenerative disease, or congenital conditions, communication between the brain and body can be severely impaired, even when cognitive function remains intact (Rousche & Normann, 1998).

Conventional assistive technologies that rely on residual peripheral muscle activity—such as EMG-controlled prosthetics or mechanical interfaces—can partially restore function. However, in cases of severe paralysis where peripheral signals are absent, direct access to neural activity is required through brain–computer interfaces (BCIs) (Rousche & Normann, 1999; Kim et al., 2009).

Non-invasive BCIs based on electroencephalography (EEG) offer relative safety but suffer from limited spatial resolution, as they record aggregated activity from large neuronal populations. This limitation restricts their ability to support precise, multi-dimensional control. To overcome this constraint, invasive intracortical microelectrode arrays were developed to record neural activity at the level of individual or small groups of neurons (Maynard et al., 1997).

Among these technologies, the Utah Intracortical Electrode Array (UEA) has emerged as a widely used platform for high-resolution cortical recording and stimulation. Its penetrating microelectrode architecture enables access to neuronal spike activity with superior spatial and temporal resolution, making it a foundational tool in experimental neuroscience and BCI research (Rousche & Normann, 1998; 1999).

Despite its advantages, the UEA faces significant challenges related to long-term performance, including biological tissue responses and material degradation. This paper reviews the design, functional principles, system integration, tissue response, and limitations of the Utah Intracortical Electrode Array, highlighting its role as both a benchmark technology and a reference point for the development of next-generation neural interfaces.

II. Utah Intracortical Electrode Array Design

The Utah Intracortical Electrode Array (UIEA) is constructed on a single silicon substrate measuring approximately 4–4.2 mm on each side (Maynard et al., 1997). From this substrate, one hundred electrically isolated needle-like microelectrodes are arranged in a regular 10 × 10 grid. Each electrode extends approximately 1.2 mm from the substrate and is chemically etched to form a sharp tip suitable for cortical penetration (Maynard et al., 1997). Electrical access to the electrodes is provided by gold contact pads on the reverse side of the array, which interface with external recording equipment (Maynard et al., 1997; Kim et al., 2009).

The fabrication process begins with cutting silicon wafers into square sections, each of which will become a single electrode. Orthogonal grooves are cut into the silicon using a diamond dicing saw, and these grooves are subsequently filled with a high-resistivity glass to electrically isolate each electrode (Maynard et al., 1997). The glass is then heated under controlled conditions and slowly cooled to remove bubbles and minimize thermal stress, preventing cracking due to temperature changes (Maynard et al., 1997). Excess glass is removed by grinding and polishing, leaving a grid-like structure in which small square silicon pillars are surrounded by insulating glass (Maynard et al., 1997).

To create the final needle-like form of the electrodes, a two-step acid etching process is employed. A “dynamic” etch reduces the thickness of the columns while maintaining their overall shape, followed by a “static” etch that sharpens the tips without significantly altering the base size or pillar height (Maynard et al., 1997). After shaping, the electrodes are coated with biocompatible metal layers (such as platinum with adhesion layers including titanium–tungsten) to allow reliable electrical interaction with neural tissue (Kim et al., 2009). Most of each electrode is covered with an insulating material, leaving only the tip exposed so it can detect neural signals (Kim et al., 2009).

This combination of precise micromachining, metallization, and insulation results in a dense and mechanically robust array capable of penetrating cortical tissue with minimal electrical interference between neighboring electrodes. The design enables the array to record activity from individual neurons or small neuronal populations while maintaining structural integrity, although long-term biological and signal stability remain important limitations (Rousche & Normann, 1998; Woeppel et al., 2021).

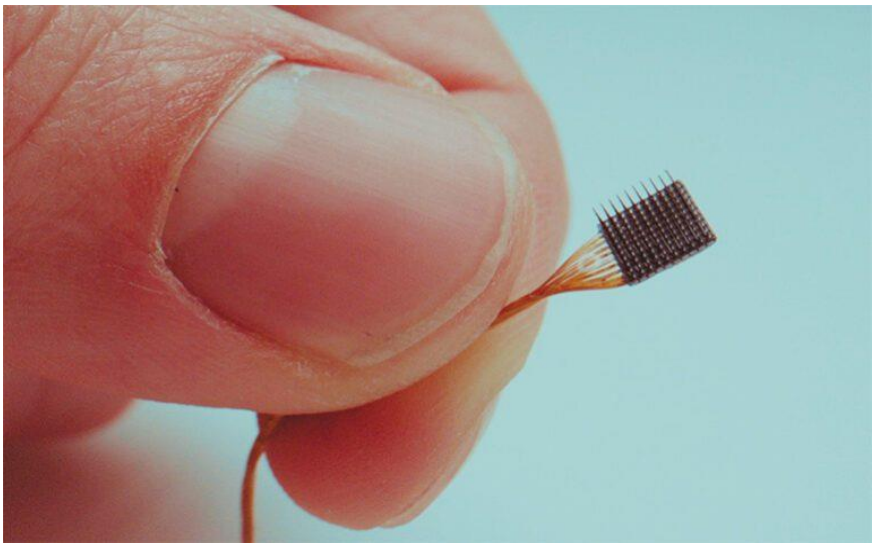


Fig 1: Blackrock Neurotech’s Utah array is used in its brain-computer interface to sense brain signals (Solzbacher, 2022)

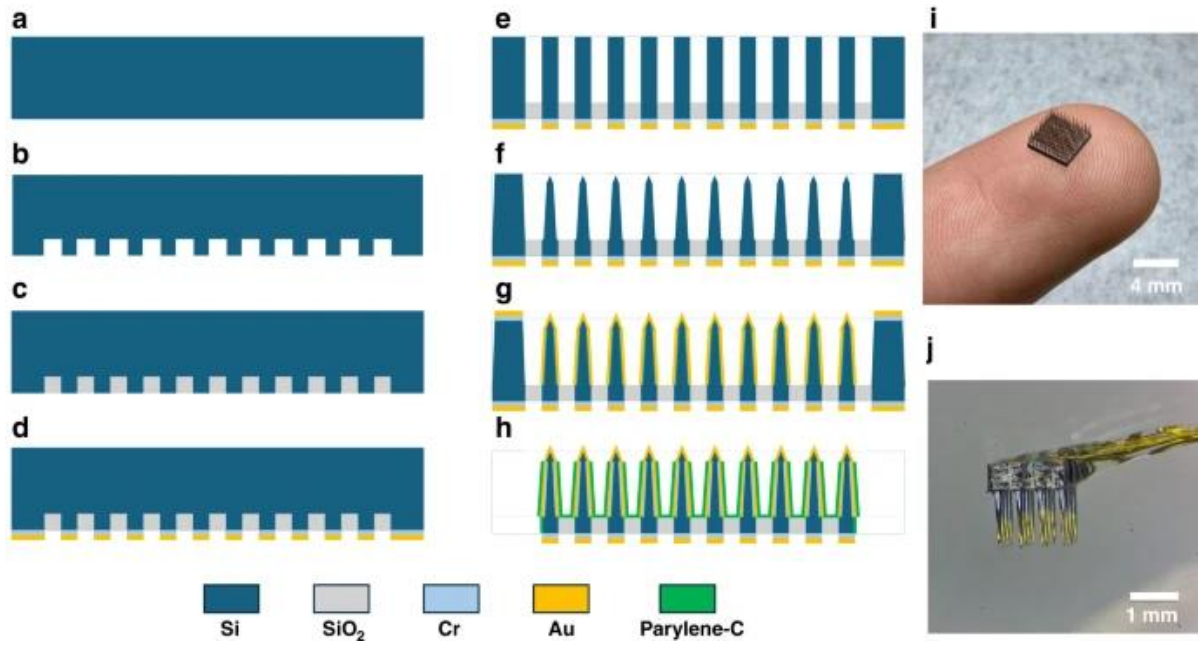


Fig. 2: Fabrication process and actual image of the silicon-based neural microelectrode array (Zhao et al., 2025).

The figure above illustrates a detailed, step-by-step fabrication process of the Utah Intracortical Electrode Array. While the top paragraph provides a high-level overview of the array's design and key manufacturing steps, this diagram breaks down the procedure into ten specific stages: (a) 1800 μm thick double-polished silicon wafer, (b) microneedle array root fabrication with etching, (c) microneedle array backboard formation with silicon–glass bonding, (d) microneedle array backend metallization, (e) microneedle array pillar formation with sawing, (f) microneedle array needle formation with wet etching, (g) microneedle array metallization, (h) local de-insulation, (i) photograph of the completed microneedle array, and (j) photograph of a packaged array. Essentially, both descriptions reflect the same overall process, but the top paragraph is a concise overview, whereas this diagram shows the detailed step-by-step fabrication.

III. Utah Intracortical Electrode Array Function

The Utah Electrode Array allows precise recording of signals from individual neurons, which are transmitted to external electronic systems for amplification and processing (Maynard et al., 1997; Kim et al., 2009). Each electrode detects electrical activity from neurons in its immediate vicinity, enabling activity detection of both individual neurons or small groups of neurons (Maynard et al., 1997). By penetrating into the cortex, the electrodes come into close proximity with neurons, providing signals with greater spatial and temporal resolution (which neuron fired and when) than non-invasive surface recordings, such as EEG (Rousche & Normann, 1998).

The penetrating design allows each electrode tip to record neuronal firing patterns with high fidelity, which is important for decoding intended motor or sensory outputs, essentially enabling the BCI to interpret a person's likely intended movements or sensations (Rousche & Normann, 1998; Kim et al., 2009). The relatively high impedance of each electrode allows it to detect the tiny voltage changes produced when neurons fire, although signal quality also depends on amplification and external recording equipment (Maynard et al., 1997).

Although the original description of the Utah Electrode Array by Maynard et al. (1997) focuses exclusively on recording performance, later studies have demonstrated that Utah-type intracortical arrays can also be used for electrical microstimulation of cortical tissue. Experimental work has shown that intracortical microstimulation delivered through such arrays can evoke neural and behavioral responses in animal models, enabling investigation of causal relationships between neural activity and function (Rousche & Normann, 1999). These later findings indicate that, while recording was the primary design objective, the Utah Electrode Array architecture is compatible with two-way (bidirectional) brain–machine interface applications (Rousche & Normann, 1999; Kim et al., 2009).

Overall, the Utah Electrode Array combines precise electrode placement, close neuronal access, and high-impedance recording capability, providing reproducible, high-resolution measurements of cortical activity and forming the basis for its extensive use in both experimental neuroscience and clinical BCI research (Maynard et al., 1997; Rousche & Normann, 1998; Kim et al., 2009).

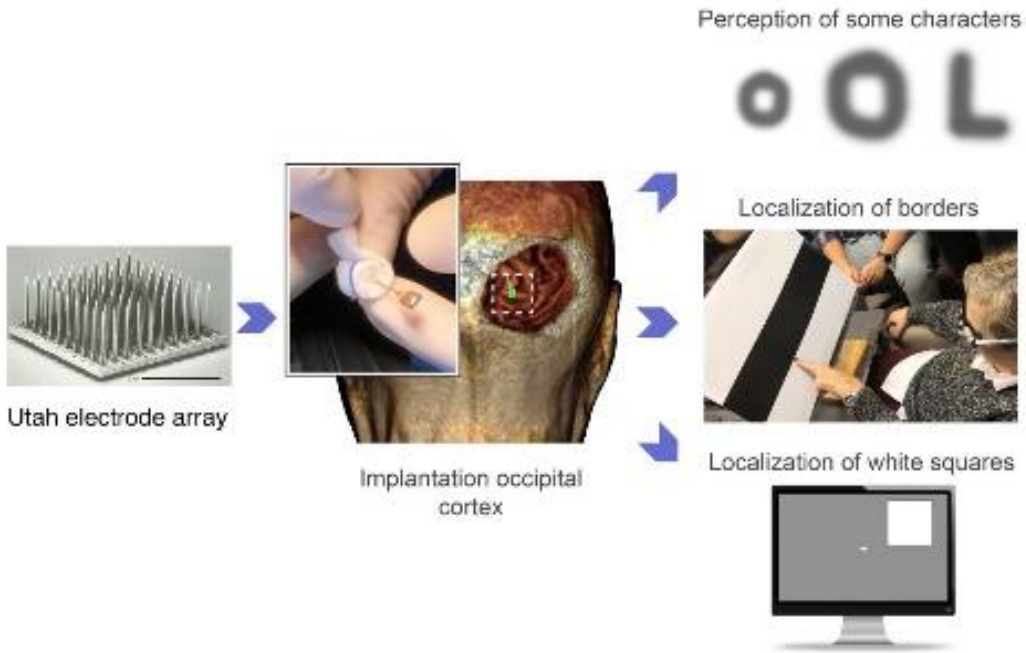


Fig 3: Brain responses to visual stimulation using an intracortical microelectrode array (Fernández et al., 2021)

IV. System Integration and Implant Packaging

The functional utility of the Utah Electrode Array (UEA) depends not only on its design and recording capabilities but also on its integration with external electronics. As described by Kim et al. (2009), effective system integration requires connecting the array to data acquisition hardware and amplification circuits. In some implementations, external systems may also incorporate wireless telemetry for signal transmission, although early UEA platforms primarily relied on wired connections. This integration allows neural activity to be monitored and processed in real time, which is important for experimental brain–computer interface applications.

Implant packaging and insulation are equally important for maintaining array performance during chronic implantation. Kim et al. (2009) describe the use of biocompatible coatings on both the array and its wires to protect the electrodes and surrounding tissue. Additionally, Maynard et al. (1997) report the use of gold contact pads on the back of the array to provide reliable electrical access to each electrode, enabling stable connections to external recording systems. Animal studies by Rousche and Normann (1998) further demonstrate that properly implanted arrays, combined with robust packaging and external interfacing, can maintain usable neural recordings over several weeks to months, although signal quality may gradually decline over longer periods.

Together, system integration and implant packaging—including biocompatible coatings, electrical connectors, and interfacing with acquisition hardware—enable the UEA to record and transmit high-fidelity neural signals in chronic experimental settings. This combination of array design, electronics, and protective measures underpins the continued importance of the UEA in experimental neuroscience and brain–computer interface research, despite known long-term limitations.

V. Tissue Response and Material Degradation

Long-term implantation of the Utah Electrode Array in the human cortex elicits both biological and material changes that can influence the quality of the signal received or sent to the neurons, even when the implant remains structurally intact and continues to record (Rousche & Normann, 1998; Woeppel et al., 2021). This decline happens because of both the body's response to the foreign material and gradual wear of the electrodes themselves.

The body reacts to implanted materials it cannot remove by initiating a biological response that includes glial scarring and fibrotic encapsulation. Glial scarring occurs when reactive brain support cells, such as astrocytes and microglia, reproduce rapidly around the electrodes, while fibrotic encapsulation involves connective tissue cells forming around the implant. Together, these processes increase the distance between the electrodes and surrounding neurons, weakening recorded signals and reducing the efficiency of the electrical interface. The amount and rate of tissue growth vary between individuals, leading to inconsistencies in signal quality over time. Although the UEA is designed to minimize acute tissue damage during implantation (Maynard et al., 1997), it cannot fully prevent these long-term cellular responses.

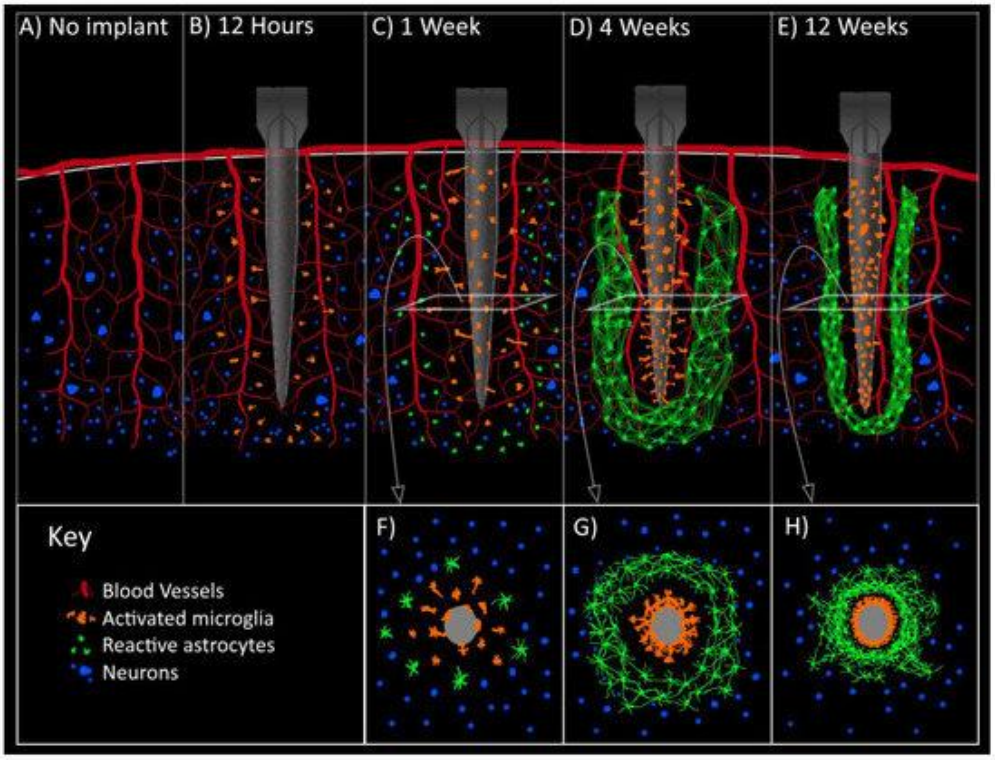


Fig 4: Illustration of glial encapsulation following neural electrode implantation. Panels (A-E) show tissue progression from pre-implantation to 12 weeks post-implantation, while panels (F-H) show cross-sectional views of the encapsulation. Although this image is not from a Utah Electrode Array implant, it demonstrates a similar biological response observed around chronically implanted intracortical electrodes.(Xie et al., 2018, Fig. 3)

Human trials also demonstrate these effects. In one clinical explant study, six Utah arrays were removed from two human participants after extended implantation periods — two arrays had been implanted for 980 days in one individual, and four arrays (two platinum and two iridium oxide) had been implanted for 182 days in another. Arrays with longer implant durations showed more pronounced tissue encapsulation, and overall signal amplitude and impedance declined over time, indicating that long-term biological responses affect recording quality in humans (Woeppel et al., 2021).

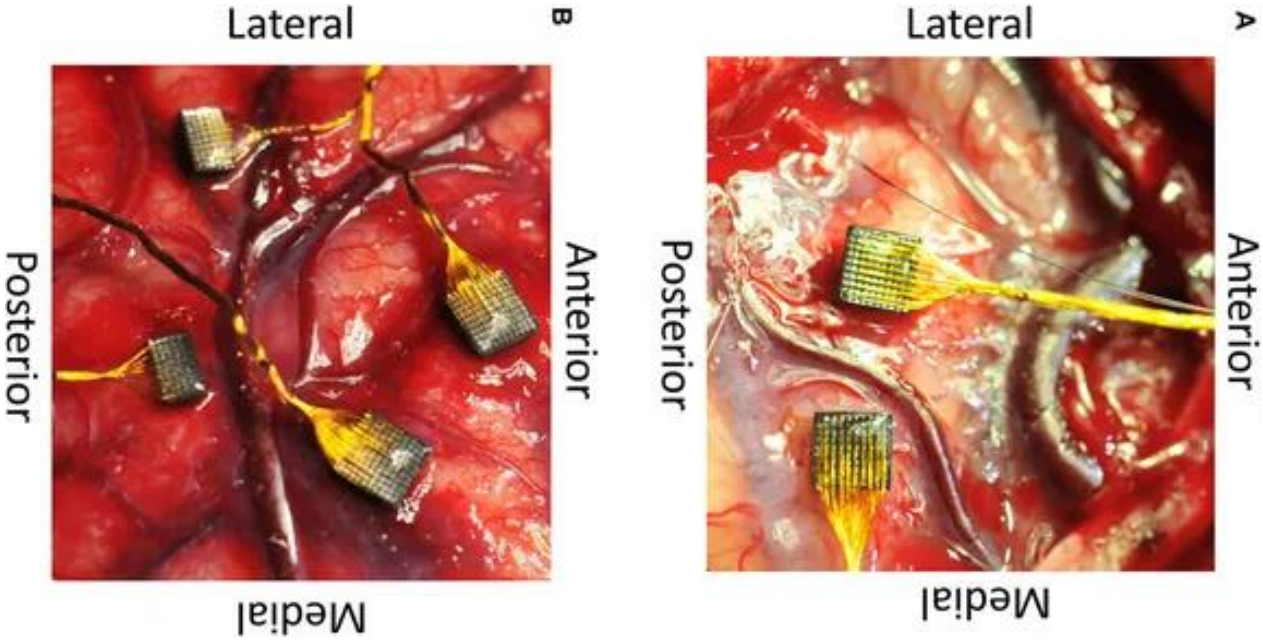


Fig 5: Utah electrode arrays implanted in two patients: Left image shows the patient with 4 arrays implanted for 182 days, Right image shows the patient with 2 arrays implanted for 980 days (Woeppel et al., 2021, Fig. 1)

The materials used in the electrodes also play a major role in how well the array performs over time. Most commonly, platinum is used for recording, and iridium oxide is used when electrical stimulation is needed. Over weeks to months, minor wear and degradation of the electrode tips can occur, metal coatings may partially delaminate, and insulation along the shafts can crack or peel. Platinum-tipped electrodes may become blunted or lose small portions of the metal layer, while iridium oxide coatings can partially delaminate at sites of repeated stimulation (Kim et al., 2009; Woeppel et al., 2021). Insulation wear can expose unintended regions of the electrode, creating electrical interference and further reducing signal stability.

Together, these biological and material factors explain why the Utah Electrode Array, while effective for high-resolution neural recording, is not permanent. Its strength lies in providing detailed neuronal information reliably for weeks to months rather than years. These observations highlight the importance of developing new materials, flexible electrode designs, and strategies to reduce inflammatory tissue responses, which will be critical for the next generation of long-term neural interface technologies.

VI. Limitations of the Utah Electrode Array

Regardless of how impactful or innovative this technology may be, especially with how transformative it was for BCI research, the UIEA still exhibits several limitations in terms of long-term performance and clinical applicability. Signal degradation over time is one of the main constraints, even when the implant remains structurally intact and continues to record (Rousche & Normann, 1998; Woeppel et al., 2021). Gradual declines in amplitude and fidelity of recorded neuronal activity can reduce reliability over extended periods.

Individual differences in biological responses contribute to variability in recording quality. While the UIEA's design minimizes acute tissue damage (Maynard et al., 1997), it cannot fully prevent the long-term reactions that affect signal consistency, such as tissue encapsulation and gliosis. Material wear also plays a role, as microelectrodes can degrade, metal coatings may flake, and insulation may crack or peel with prolonged use (Kim et al., 2009; Woeppel et al., 2021). These factors together illustrate why the array, while effective for high-resolution recordings, is not permanent and is best suited for providing detailed neuronal information for weeks to months rather than years.

Understanding these limitations is essential for setting realistic expectations in experimental and clinical applications, and for guiding the development of next-generation neural interfaces with improved materials, flexible designs, and strategies to reduce inflammatory tissue responses.

VII. Conclusion

The Utah Electrode Array represents a landmark technology in neural interfacing, providing high-resolution access to cortical neuronal activity for recording (Maynard et al., 1997; Rousche & Normann, 1998) and, in experimental settings, for intracortical microstimulation (Rousche & Normann, 1999). Its penetrating microelectrode design enables single-unit and multi-unit recordings with spatial and temporal resolution unattainable with non-invasive methods, making it a powerful tool for experimental neuroscience and brain-computer interface applications.

Long-term studies in both animal models and human participants demonstrate that the UEA is capable of chronic neural recording, maintaining functional signals over weeks to months. However, these studies also reveal gradual declines in signal quality, driven by biological responses such as glial scarring and fibrotic encapsulation, as well as material degradation including coating delamination and insulation failure (Rousche & Normann, 1998; Woeppel et al., 2021; Kim et al., 2009).

The interaction between the electrode materials and host tissue is a key factor limiting long-term performance. Despite these constraints, the UEA continues to provide invaluable insights into neuronal dynamics and serves as a benchmark for the development of next-generation intracortical arrays. Its combination of precise design, high-resolution functionality, and intermediate-term applicability ensures that it remains a cornerstone technology in the study and application of neural interfaces.

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